

Angle Bounded Linear Operators

Definition: $K: X \rightarrow X^*$ is angle bounded with constant $c \geq 0$ if:

$$|(Kx_1, x_2) - (Kx_2, x_1)| \leq 2c \sqrt{(Kx_1, x_1)(Kx_2, x_2)}$$

Properties:

- Monotone: $(Kx, x) \geq 0$ for all $x \in X$
- Symmetric: $K^* = K$

Examples: sector: numerical range in a sector

- Linear differential operators
- Important for: variational methods, PDE theory
- Convolution operators
- Integral operators with symmetric kernels